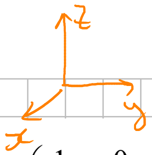
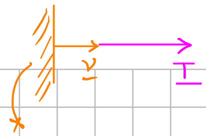


$$(\sigma_{ij}) = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

Fórmula de Cauchy $\underline{T}^{\nu(i)} = \underline{\underline{\sigma}}\,\underline{\nu}^{(i)}$

$\underline{T}^{\nu(i)} = \lambda^{(i)} \cdot \underline{\nu}^{(i)}$



$$\left\{ \begin{array}{l} \underline{\underline{\sigma}}\,\underline{\nu}^{(i)} = \lambda\,\underline{\nu}^{(i)} \\ (\underline{T} \,//\, \underline{\nu}^{(i)}) \end{array} \right.$$

$$\underline{\underline{\sigma}}\,\underline{\nu}^{(i)} - \lambda^{(i)}\,\underline{\nu}^{(i)} = 0$$

$$\sigma_{kj}(\nu^{(i)})_j - \lambda^{(i)} \cdot \underbrace{(\nu^{(i)})_k}_{\delta_{kj}(\nu^{(i)})_j} = 0$$

$$\underline{\underline{\sigma}}\,\underline{\nu}^{(i)} - \lambda^{(i)}\underline{\underline{I}}\,\underline{\nu}^{(i)} = 0$$

$$\underbrace{(\underline{\underline{\sigma}} - \lambda^{(i)}\underline{\underline{I}})}_{\underline{\underline{A}}}\,\underline{\nu}^{(i)} = 0$$

$\Rightarrow \det(\underline{\underline{A}}) = 0 \rightarrow$ Ecuación característica

↳ singular para que no resulte solo en la solución trivial de $\underline{\nu}^{(i)}$

$$|\tau_{ij} - \sigma\delta_{ij}| = \begin{vmatrix} \tau_{11} - \cancel{\sigma}\,\lambda^{(i)} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} - \cancel{\sigma}\,\lambda^{(i)} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} - \cancel{\sigma}\,\lambda^{(i)} \end{vmatrix}$$

$$= -\sigma^3 + I_1\sigma^2 - I_2\sigma + I_3 = 0,$$

$\det(B) = \det(B^T)$

$\sigma_{33}\sigma_{11} - \sigma_{31}\sigma_{13} = \sigma_{11}\sigma_{33} - \sigma_{13}\sigma_{33}$

$$\sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

$$I_1 = \tau_{11} + \tau_{22} + \tau_{33},$$

$\rightarrow I_1(\underline{\underline{\sigma}}) = \mathrm{tr}(\underline{\underline{\sigma}})$ traza → suma de los componentes de la diagonal de $\underline{\underline{\sigma}}$

$$I_2 = \begin{vmatrix} \tau_{22} & \tau_{23} \\ \tau_{32} & \tau_{33} \end{vmatrix} + \begin{vmatrix} \tau_{33} & \tau_{31} \\ \tau_{13} & \tau_{11} \end{vmatrix} + \begin{vmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{vmatrix},$$

$$I_2(\underline{\underline{\sigma}}) = \frac{1}{2}\left[\underbrace{\mathrm{tr}^2(\underline{\underline{\sigma}})}_{I_1(\sigma)^2} - \mathrm{tr}(\underline{\underline{\sigma}}^2)\right] \qquad \underline{\underline{\sigma}} \cdot \underline{\underline{\sigma}}$$

$$\hookrightarrow -\begin{vmatrix} \sigma_{31} & \sigma_{33} \\ \sigma_{11} & \sigma_{13} \end{vmatrix} = +\begin{vmatrix} \sigma_{11} & \sigma_{13} \\ \sigma_{31} & \sigma_{33} \end{vmatrix}$$

$$I_3 = \begin{vmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{vmatrix}.$$

$\rightarrow I_3(\underline{\underline{\sigma}}) = \det(\underline{\underline{\sigma}})$

$(-1-\lambda) \cdot \lambda(-2+\lambda) \longrightarrow \lambda = -1,\ \lambda = 0,\ \lambda = 2$

$-2\lambda + \lambda^2$

$\cancel{1} - 2\lambda + \lambda^2 \cancel{-1}$

$(1-\lambda)^2 - 1$

$(-1-\lambda) \cdot [(1-\lambda) \cdot (1-\lambda) - 1]$

$\lambda_1 > \lambda_2 > \lambda_3$

$$\begin{vmatrix} 1-\lambda & 0 & -1 \\ 0 & -1-\lambda & 0 \\ -1 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda) \cdot (-1-\lambda) \cdot (1-\lambda) - (-1-\lambda)$$

$$= \lambda(\underbrace{-\lambda^2 + \lambda + 2}_{}) \rightarrow \lambda = 0 = \lambda_2$$

Cuadrática → $\lambda = -1 = \lambda_3$; $\lambda = 2 = \lambda_1$

$$\det(A) = (-1)^{2+2} \cdot (-1-\lambda) \cdot \begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} = +1 - (1+\lambda) \cdot [(1-\lambda)^2 - 1]$$

↳ $\lambda = -1$

↳ $\lambda^{(3)} = -1$

→ $(1-\lambda)^2 - 1 = 0$

$\cancel{1} - 2\lambda + \lambda^2 \cancel{-1} =$

$\lambda(-2+\lambda) = 0$

↳ $\lambda = 0$ → $\lambda = 2$

↳ $\lambda^{(2)} = 0$ ↳ $\lambda^{(1)} = 1$

$$\left(\underline{\underline{T}} - \lambda^{(i)} \underline{\underline{I}}\right) \underline{v}^{(i)} = 0 \qquad \text{incógnita} \qquad \|v^{(i)}\| = \sqrt{v_1^{(i)2} + v_2^{(i)2} + v_3^{(i)2}} = 1$$

$$\begin{bmatrix} 1-\lambda^{(i)} & 0 & -1 \\ 0 & -1-\lambda^{(i)} & 0 \\ -1 & 0 & 1-\lambda^{(i)} \end{bmatrix} \cdot \begin{bmatrix} v_1^{(i)} \\ v_2^{(i)} \\ v_3^{(i)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} (1-\lambda^{(i)}) \cdot v_1^{(i)} - v_3^{(i)} = 0 \\ (-1-\lambda^{(i)})\, v_2^{(i)} = 0 \qquad \Rightarrow \lambda^{(i)} \neq -1 \Rightarrow v_2^{(i)} = 0 \\ -1\, v_1^{(i)} + (1-\lambda^{(i)})\, v_3^{(i)} = 0 \end{cases}$$

- $\lambda^{(1)} = 2 \Rightarrow \begin{cases} v_2^{(1)} = 0 \\ -v_1^{(1)} - v_3^{(1)} = 0 \\ -v_1^{(1)} - v_3^{(1)} = 0 \end{cases} \Rightarrow v_1^{(1)} = -v_3^{(1)}$ $\qquad v_3^{(1)} = a,\ v_1 = -a$

$$\|\underline{v}^{(1)}\| = \sqrt{(-v_3^{(1)})^2 + 0^2 + (v_3^{(1)})^2} = 1 \Rightarrow \sqrt{2}\, |v_3^{(1)}| = 1 \Rightarrow v_3^{(1)} = \pm 1/\sqrt{2}$$

$$\Downarrow$$

$$v_1^{(1)} = \mp 1/\sqrt{2}$$

- $\lambda^{(2)} = 0 \Rightarrow \begin{cases} v_2^{(2)} = 0 \\ v_1^{(2)} - v_3^{(2)} = 0 \\ -v_1^{(2)} + v_3^{(2)} = 0 \end{cases} \quad v_1^{(2)} = v_3^{(2)}$

$$\|\underline{v}^{(2)}\| = 1 \Rightarrow v_3^{(2)} = \pm \frac{1}{\sqrt{2}} \Rightarrow v_1^{(2)} = \pm \frac{1}{\sqrt{2}}$$

- $\lambda^{(3)} = -1 \Rightarrow 2 v_1^{(3)} - v_3^{(3)} = 0 \Rightarrow v_3^{(3)} = 2 v_1^{(3)}$

$0 \cdot v_2^{(3)} = 0 \Rightarrow v_2^{(3)} \in \mathbb{R}$ (tensor $\underline{\underline{T}}$ simétrico)

$-v_1^{(3)} + 2 v_3^{(3)} = 0 \Rightarrow -v_1^{(3)} + 2 \cdot 2 v_1^{(3)} = 0 \Rightarrow v_1^{(3)} = 0 \Rightarrow$

$3 v_1^{(3)} = 0$

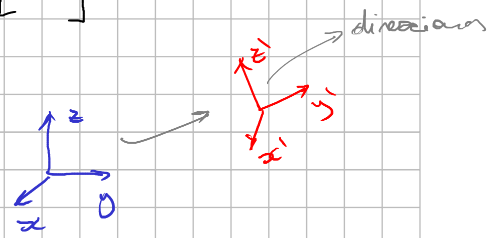
$v_3^{(3)} = 0$

$$\|\underline{v}^{(3)}\| = 1 \Rightarrow |v_2^{(3)}| = 1 \Rightarrow v_2^{(3)} = \pm 1$$

Autovectores o direcciones principales:

$$\underbrace{\underline{v}^{(1)}}_{\underline{e}'_1} = \begin{bmatrix} -\alpha \\ 0 \\ \alpha \end{bmatrix}, \quad \underbrace{\underline{v}^{(2)}}_{\underline{e}'_2} = \begin{bmatrix} \alpha \\ 0 \\ \alpha \end{bmatrix}, \quad \underbrace{\underline{v}^{(3)}}_{\underline{e}'_3} = \begin{bmatrix} 0 \\ \gamma \\ 0 \end{bmatrix}$$

$$\alpha = \pm \frac{1}{\sqrt{2}} \qquad \gamma = \pm 1$$



$$\{x\} \to \{x'\}$$

fila   columna

$$B_{ij} = \underline{e}'_i \cdot \underline{e}_j = (\underline{e}'_i)_j = \begin{bmatrix} [(e'_1)_1 \ (e'_1)_2 \ (e'_1)_3] \\ [\quad \underline{e}'_2 \quad] \\ [\quad \underline{e}'_3 \quad] \end{bmatrix}$$

$$\underline{\underline{B}} = \begin{bmatrix} -\alpha & 0 & \alpha \\ \alpha & 0 & \alpha \\ 0 & \gamma & 0 \end{bmatrix} \Rightarrow \det(\underline{\underline{B}}) = (-1)^{3+2} \cdot \gamma \cdot \begin{vmatrix} -\alpha & \alpha \\ \alpha & \alpha \end{vmatrix}$$

$$= -\gamma \cdot (-\alpha^2 - \alpha^2) = 2\gamma\alpha^2$$

$\rightarrow \underline{e}'_3 = \underline{e}'_1 \times \underline{e}'_2$ Como $\alpha \in \mathbb{R} \Rightarrow$

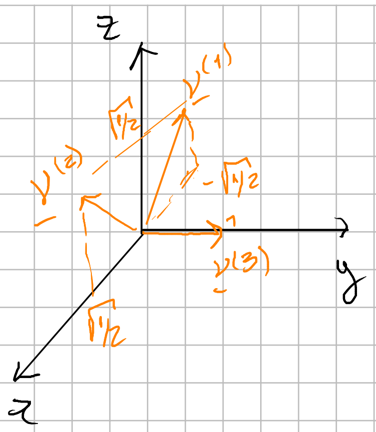
$$\det(\underline{\underline{B}}) = 1 = 2\gamma\alpha^2 \Rightarrow |\alpha| = \frac{1}{\sqrt{2\gamma}} \Rightarrow \gamma \in \mathbb{R} \Rightarrow \gamma = +1$$

$\gamma < 1$

dextrógira

$$|\alpha| = \frac{1}{\sqrt{2}} \Rightarrow \alpha = \pm\sqrt{\frac{1}{2}}$$

$$\alpha = +\sqrt{\frac{1}{2}} \Rightarrow \underline{v}^{(1)} = \begin{bmatrix} -\sqrt{1/2} \\ 0 \\ \sqrt{1/2} \end{bmatrix}, \quad \underline{v}^{(2)} = \begin{bmatrix} \sqrt{1/2} \\ 0 \\ \sqrt{1/2} \end{bmatrix}, \quad v^{(3)} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$



$\underline{\nu}^{(3)} = \underline{\nu}^{(1)} \times \underline{\nu}^{(2)}$

$$\det(\underline{\underline{\beta}}) = -1 \Rightarrow |\alpha| = \frac{1}{\sqrt{-2\gamma}} \Rightarrow \gamma = -1$$

levógira

$$\Rightarrow \alpha = \pm\sqrt{\frac{1}{2}}$$

—o—

$$(\sigma_{ij}) = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

$I_1 = 1 - 1 + 1 = 1$ $\qquad (1\times 1) - (-1 \times 1)$

$$I_2 = \begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} = -1 + 0 - 1 = -2$$

$$I_3 = \begin{vmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = -1 - (-1) = 0$$

$$-\lambda^3 + 1\cdot\lambda^2 - (-2)\cdot\lambda + 0 = 0$$

$$-\lambda(\lambda^2 - \lambda - 2) = 0$$

$\lambda = 0$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{1 \pm \sqrt{(-1)^2 - 4\cdot 1\cdot(-2)}}{2\cdot 1} =$$

$$= \frac{1 \pm \sqrt{9}}{2} = \frac{1 \pm 3}{2} = \begin{cases} \lambda = 2 \\ \lambda = -1 \end{cases}$$